Last time: For a graph G,

- A proper coloring is a function $f\colon V(G) \to C$ where $f(u) \neq f(v)$ if $uv \in E(G)$.
- A proper coloring using t colors corresponds to a t partition of G.
- The chromatic number $\chi(G)$ is the smallest number of colors in any proper coloring.
- Lower bounds $\chi(G) \geqslant \omega(G)$ and $\chi(G) \geqslant \frac{n}{\alpha(G)}$. $n = |V(G)|$

  The *clique number* $\omega(G)$ is the size of the largest clique contained in G.
- Upper bounds $\chi(G) \leqslant d(G) + 1 \leqslant \Delta(G) + 1$.

  The *degeneracy* of G is $d(G) = \max\{\delta(H)\colon H \text{ is a subgraph of } G\}$.

Next: The bound $\chi(G) \geqslant \omega(G)$ is very loose.

We say that G is **triangle-free** if it contains no copies of K_3. Equivalently, $\omega(G) \leqslant 2$.

$\omega = 1$ (two edges)

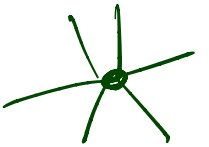
$\omega = 2$ $E(G) \neq \emptyset$



Definition. For a graph G with $V(G) = \{v_1, \dots, v_n\}$, the **Mycielski graph** $\mu(G)$ is obtained from G by adding:

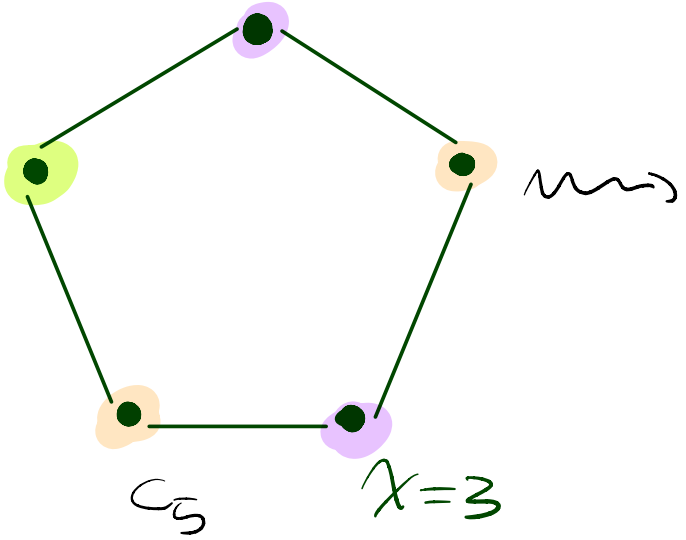
(1) *shadow vertices* $\{u_1, \dots, u_n\}$ and edges $u_i v_j \in E(\mu(G))$ if $v_i v_j \in E(G)$,

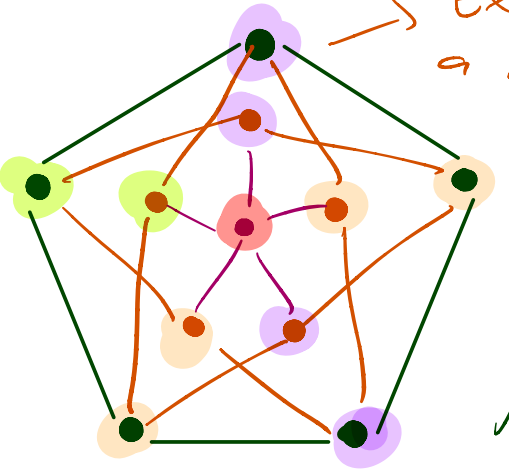
(2) a vertex w and edges wu_i for all shadow vertices u_i.



→ star vertex

Example (The Grötzsch graph $\mu(C_5)$).





→ Exercise: Show a 4-proper coloring

Grötzch graph.

$\mu(\ ) = ?$

$= 3+1$

$= 4$

lemma 2.

Theorem (10.10). *For every integer* $k \geqslant 3$, *there exists a triangle-free graph* G *with* $\chi(G) = k$.

$\omega(G) \leq 2$

Lemma 1. *If* H *is triangle-free, then* $\mu(H)$ *is triangle free.*

Lemma 2. $\chi(\mu(H)) = \chi(H) + 1$.

HW11

$\frac{\omega(G)}{2} \leq \chi(G)$, k, e.g. 10^{100}

Idea: $k = \chi(H)$. If we have a proper coloring $f: V(H) \to \{1 \ldots k\}$

Desire proper coloring on $\mu(G)$: shadow v_i of v_i can take same color, for star w, use new color.

Thus have $(k+1)$ proper coloring for $\mu(H)$.

If $\mu(H)$ has k-proper coloring, the star vertex w uses some color, k, and all shadow vertices take colors in $\{1, \ldots, k-1\}$

Use that to get $(k-1)$-proper coloring of H.

Theorem (10.10). *For every integer* $k \geqslant 3$*, there exists a triangle-free graph* G *with* $\chi(G) = k$.

Lemma 1. *If* H *is triangle-free, then* $\mu(H)$ *is triangle free.*

Lemma 2. $\chi(\mu(H)) = \chi(H) + 1$.

Proof: Induction. Base case $k=3$, Take $G = C_5$

If H is Triangle free with $\chi(H) = k$

Then $\mu(H)$ is a Triangle free (lemma 1)

with $\chi(\mu(H)) = k+1$ (lemma 2). ∎

$\omega(G) \leq \chi(G)$

Theorem. *If G is an n-vertex graph, then $\chi(G) \cdot \chi(\overline{G}) \geqslant n$.*

Lemma. $\omega(\overline{G}) = \alpha(G)$.

Proof: Let $V = V(G) = V(\overline{G})$. Take $A \subseteq V$

Then A is independent set for $G \iff \forall u \neq v \in A,\ uv \notin E(G)$

$\iff \forall u \neq v \in A \quad uv \in E(\overline{G})$

$\iff A$ is complete in $\overline{G}$

$\overline{G}[A]$ (induced)

Thus $\alpha(G)$ = maximum size of independent set for G

= maximum size of a clique in $\overline{G}$ ☑ Lemma

= $\omega(\overline{G})$

Either $\chi(G) \geqslant \sqrt{n}$ or $\chi(\overline{G}) \geqslant \sqrt{n}$

Proof Thm: $\chi(G)\,\chi(\overline{G}) \geqslant \frac{n}{\alpha(G)} \cdot \omega(\overline{G}) = n.$ ☑ Thm.

Theorem. *If* G *is an* n*-vertex graph, then* $\chi(G) \cdot \chi(\overline{G}) \geqslant n$.

[redacted].

Second proof: (HW11) Let $V = V(G) = V(\overline{G})$

Take $f_1 : V \longrightarrow C_1$ proper coloring for G, with $|C_1| = \chi(G)$

$f_2 : V \longrightarrow C_2$ proper coloring for $\overline{G}$, with $|C_2| = \chi(\overline{G})$

Define $f : V \longrightarrow C_1 \times C_2$, $f(x) = (f_1(x), f_2(x))$

Claim: If $x \neq y$ Then $f(x) \neq f(y)$

Case 1: $xy \in E(G) \Rightarrow f_1(x) \neq f_1(y) \Rightarrow f(x) \neq f(y)$

Case 2: $xy \notin E(G) \Rightarrow xy \in E(\overline{G}) \Rightarrow f_2(x) \neq f_2(y)$
$\Rightarrow f(x) \neq f(y)$

Thus f is 1-1, Hence $\underbrace{|V|}_{n} \leq |C_1 \times C_2| = \underbrace{|C_1|}_{\chi(G)} \times \underbrace{|C_2|}_{\chi(\bar{G})}$

∎

Definition. *Let* t *be a positive integer. A graph* G *is* t**-critical** *if:*

(1) $\chi(G) \geqslant t$, *and*

(2) $\chi(H) \leqslant t-1$ *for any proper subgraph* H *of* G.

Some observations:

(1) The only 1-critical graph is K_1. ✓

(2) The only 2-critical graph is K_2.

K_2 is 2-critical ✓

If G is 2-critical, Then the subgraphs are isolated vertices

(3) G is 3-critical if and only if G is an odd cycle.

HW11

4) No classification of 4-critical graphs

Proposition 3. *If a graph G has $\chi(G) \geq t$, then it contains a subgraph which is t-critical.*

Proof: Let $\mathcal{F} = \{$all subgraphs H of G with $\chi(H) \geq t\}$

Note: $\mathcal{F} \neq \emptyset$, because $G \in \mathcal{F}$.

Hence, among all elements in $\mathcal{F}$, there is one, say H, which minimizes $|V(H)| + |E(H)|$

Claim: H is t-critical.

1) $\chi(H) \geq t$, because $H \in \mathcal{F}$.

2) If H' is any proper subgraph of H, then

$$|V(H')| + |E(H')| < |V(H)| + |E(H)|$$

Thus H' is not in $\mathcal{F}$, hence $\chi(H') \leq t-1$ ∎

Proposition 4. *If* G *is* t*-critical, then* $\chi(G) = t$ *and* $\delta(G) \geqslant t - 1$.

Proof: For $t=1$, $G = K_1$, $\chi(G) = 1$, $\delta(G) = 0$

Assume $t \geqslant 2$. Since $\chi(G) \geqslant t \geqslant 2$, G has more than one vertex

Fix $v \in V(G)$ such that $\deg v = \delta(G)$. Consider

$H = G - v$ which is a proper subgraph of G

Since G is t-critical, $\chi(H) \leq t-1$

Let $f : V(H) \longrightarrow \{1, \ldots, t-1\}$ proper coloring for H.

$\chi(G) = t$ | Extend f to $V(G)$ by defining $f(v) = t$

This coloring for G is proper because the color t was un-used by $V(H)$. Thus $\chi(G) \leq t$

But $\chi(G) \geqslant t$ (by def^n) Hence $\chi(G) = t$.

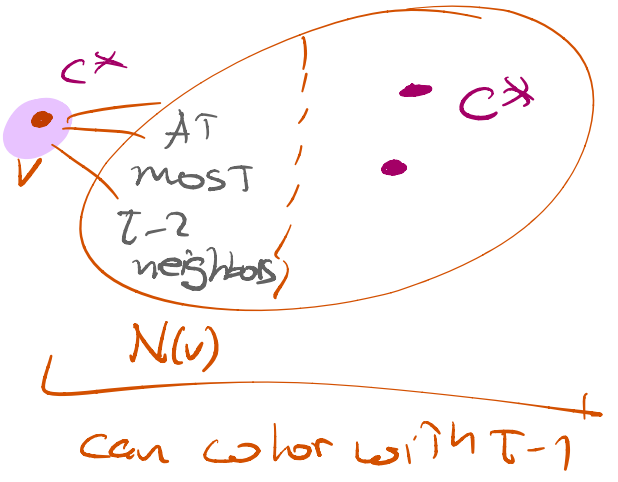
$\delta(G) \geq \tau - 1$ Assume for a contradiction $\delta(G) \leq \tau - 2$

Thus $\deg v = \delta(G) \leq \tau - 2$.

Then under $f : V(H) \longrightarrow \{1, \ldots, \tau - 1\}$ at least one color, $c^* \in$ is not used by neighbors of v

Extend f to $V(G)$ by defining $f(v) = c^*$. Since c^* is not used by $N(v)$, this is a proper coloring $V(G) \longrightarrow \{1, \ldots, \tau - 1\}$.

Contradicts $\chi(G) = \tau$.



Recall

Theorem (4.9). *Let T be a tree on t-vertices. If G is any graph with $\delta(G) \geqslant t-1$, then G contains a copy of T.*

Now we can also show:

Theorem. *Let T be a tree on t-vertices. If G is any graph with $\chi(G) \geqslant t$, then G contains a copy of T.*

Proof: Since $\chi(G) \geqslant T$

Prop 3 $\Rightarrow$ G contains a t-critical graph H.

Prop 4 $\rightarrow$ $\delta(H) \geqslant t-1$

Thm 4.9 $\Rightarrow$ H contains a copy of T.

$\Rightarrow$ G contains a copy of T.

